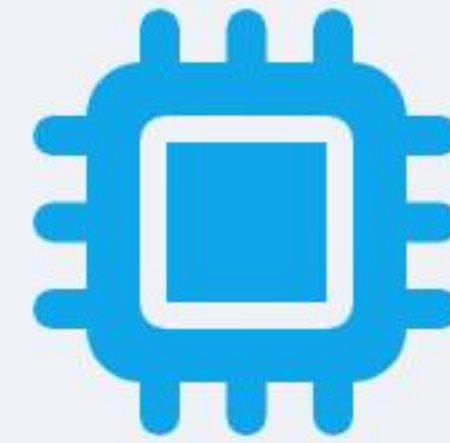




Delta RL-Guided Memetic Algorithm for CVRP

Research Project: Optimizing Logistics with Artificial Intelligence

Author 1 & Author 2

Indian Institute of Information Technology, Allahabad

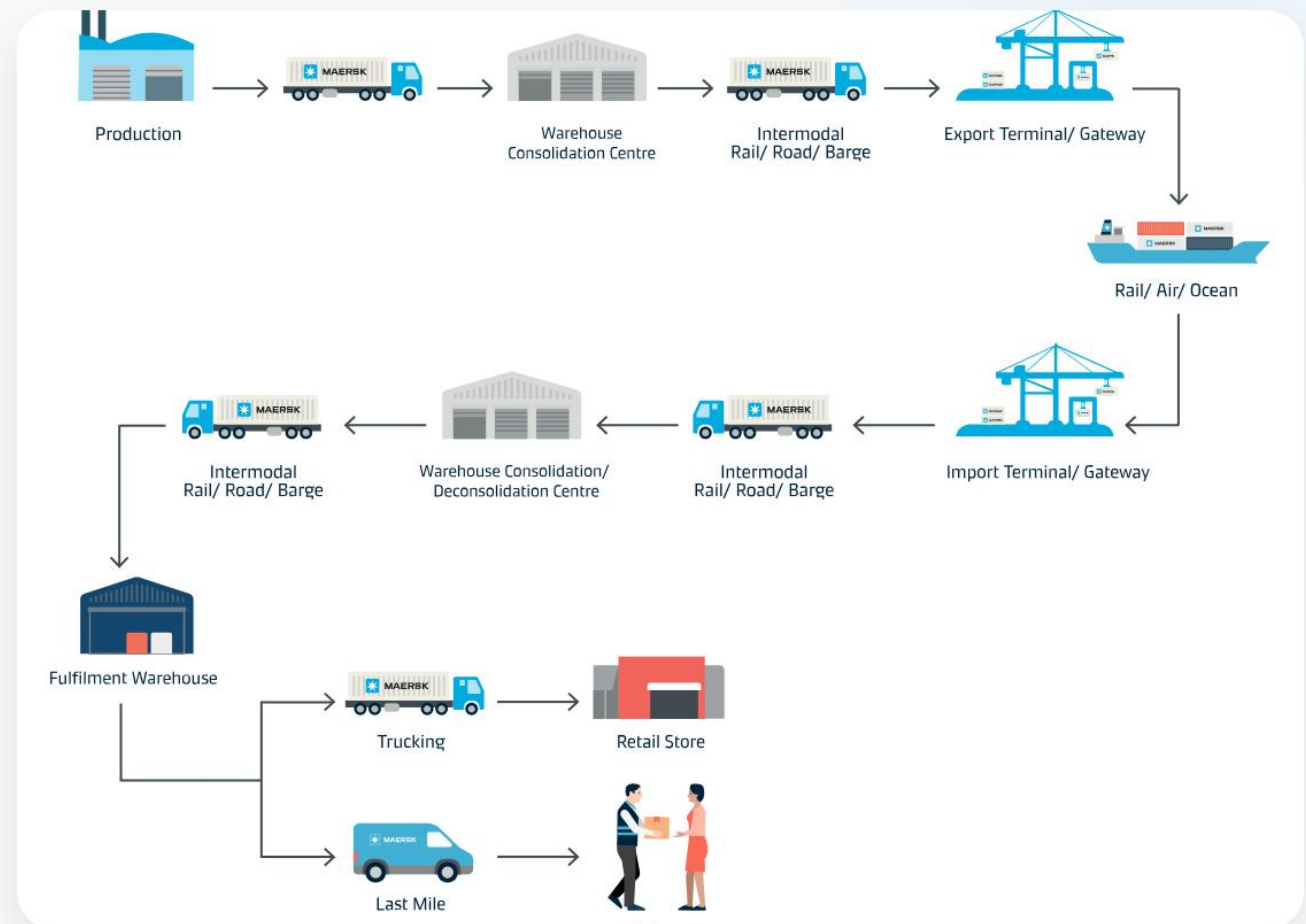
What is the CVRP Problem?

Delivery Logistics Simplified

Imagine a warehouse (Depot) with delivery trucks. Each truck has a **maximum weight capacity**.

- You have many customers to visit once.
- The goal is to find the **shortest path**.
- No truck should be overloaded.

It is "NP-Hard," meaning it's incredibly difficult to solve as the number of customers grows.



| What is a Genetic Algorithm (GA)?



Population

Start with 100 random delivery route plans (solutions).



Selection

Pick the "fittest" plans—those with the shortest travel distance.



Evolution

Mix parts of good plans (Crossover) or tweak them slightly (Mutation) to find even better ones.

The Problem with Traditional GA

Static Blindness

Regular GAs use the same mutation rate throughout the search. They don't "learn" which tool is working best right now.

Result: They get stuck in "Local Minima"—good solutions, but not the *best* ones.

The RL Advantage

Reinforcement Learning acts as a **Coach**. It watches the population and says: "Hey, we are stuck! Try a bigger change (Block Swap) instead of a small tweak."

It adapts to the problem dynamically.

| Base Paper: Quevedo et al. (2021)

RL-GA Framework

This paper used Q-Learning to tune **Crossover and Mutation Rates** (e.g., set mutation to 5% or 10%).

- 30 states (simple environment).
- 100 possible actions.
- Achieved up to 11% improvement over standard algorithms.

Our work builds on this by controlling the **Operator Type** directly.

 RL Agent Loop

Proposed Approach: Delta RL-MA

A "Memetic" algorithm combines Genetic Algorithms with Local Search (Hill Climbing). We guide this using a Delta-RL agent.

System Architecture

Our agent observes **6 population features** (Total Cost, Variance, etc.).

It then chooses an operator. All 100 routes then try this operator 25 times to improve (Deep Local Search).

The **Delta Reward** is calculated based on how much the total cost improved.

| How we represent a Route?

We use **Integer Encoding**. Think of numbers as customer ID names.

0 | 3 | 5 | 2 | 0 | 1 | 4 | 0

0 = Depot (Warehouse). Other numbers are Customers.

Route 1: Start at 0 → 3 → 5 → 2 → Back to 0.

Route 2: Start at 0 → 1 → 4 → Back to 0.

Small Tweaks: Swap & Relocate



Swap

Pick two customers and switch their places.

$[3, 5] \rightarrow [5, 3]$



Relocate

Move one customer from one vehicle to a completely different one.

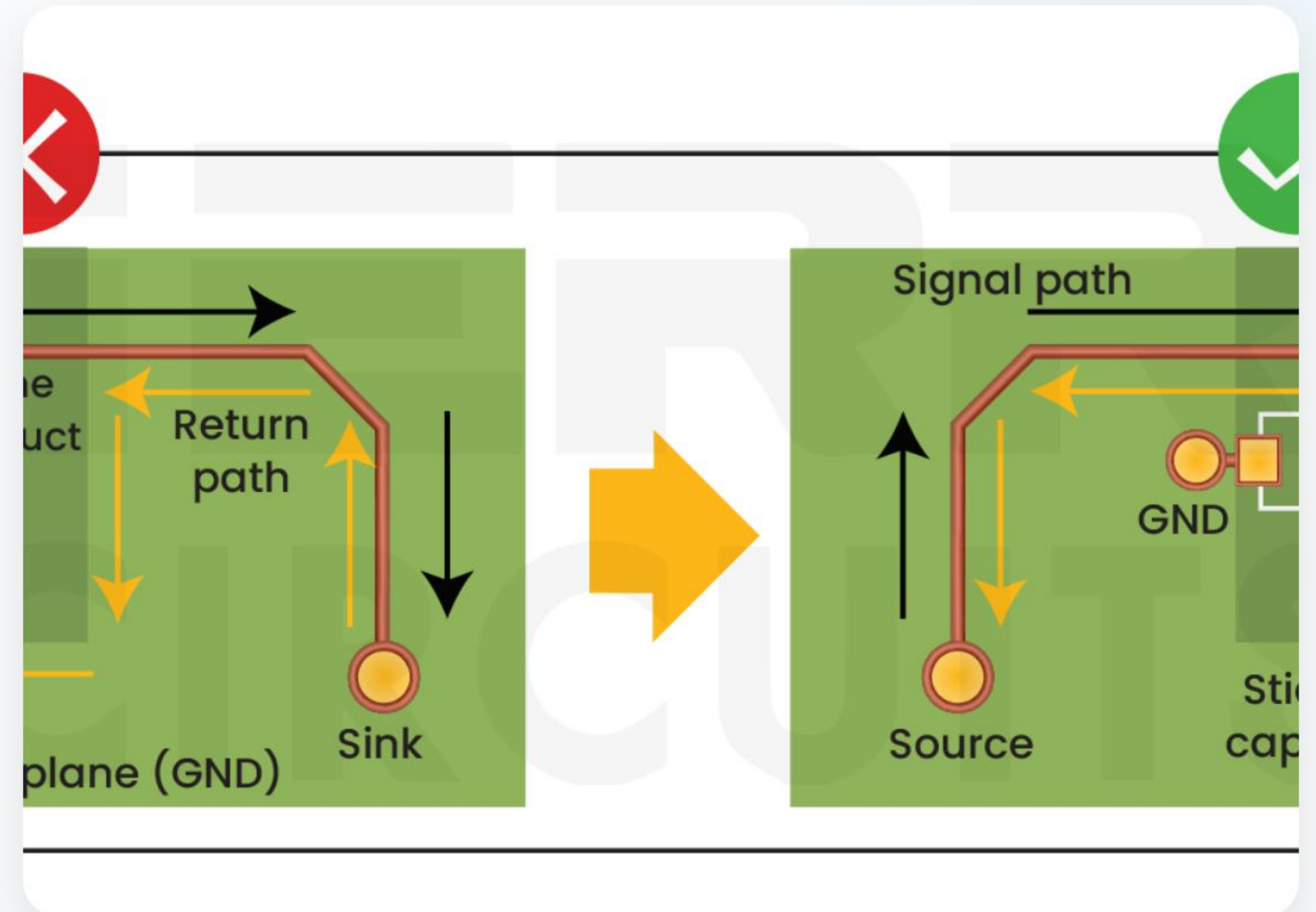
$[3, 5, 2] \mid [1, 4] \rightarrow [3, 2] \mid [1, 4, 5]$

| Advanced: 2-Opt & Block Moves

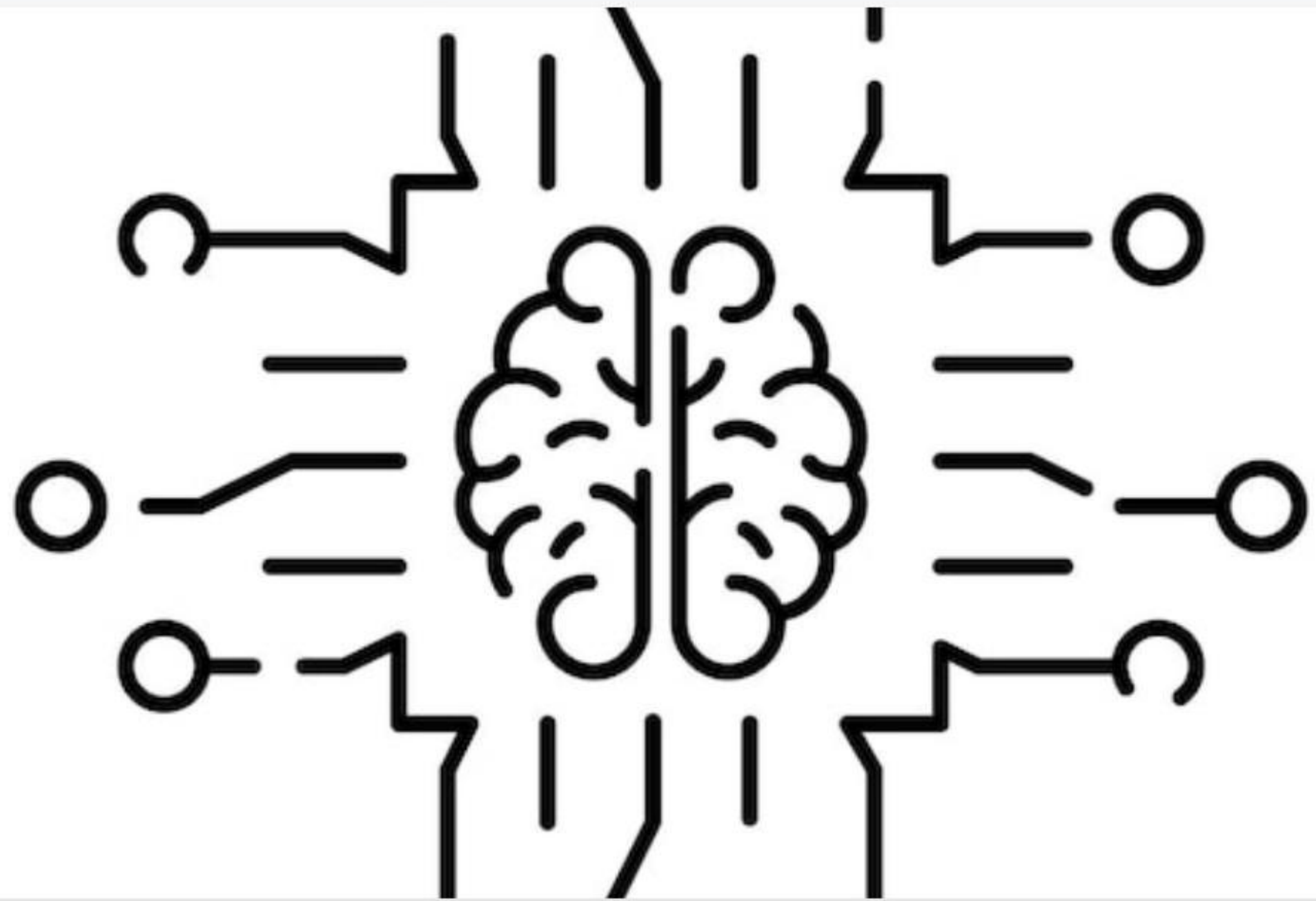
Untangling Paths

- **2-Opt:** Reverses a segment to untie "crossed" paths.
- **Block Swap:** Swaps segments of up to 5 customers at once.
- **Block Relocate:** Moves a whole group of customers.

These are the "**Heavy Hammers**" used when small swaps don't work.



| The Brain: Q-Learning Agent



States and Rewards

The agent learns by maximizing its **Reward**.

$$R = (C_{\text{old}} - C_{\text{new}}) - \lambda \cdot \text{Violation}$$

Actions: Pick one of the 5 tools.

Learning: It updates its memory (Q-Table) after every generation.

| States and Action Space

729
Discrete States

Population Feature	Bins
Total Population Cost	Low / Med / High
Route Length Variance	Low / Med / High
Capacity Violations	Low / Med / High
Average Number of Routes	Low / Med / High

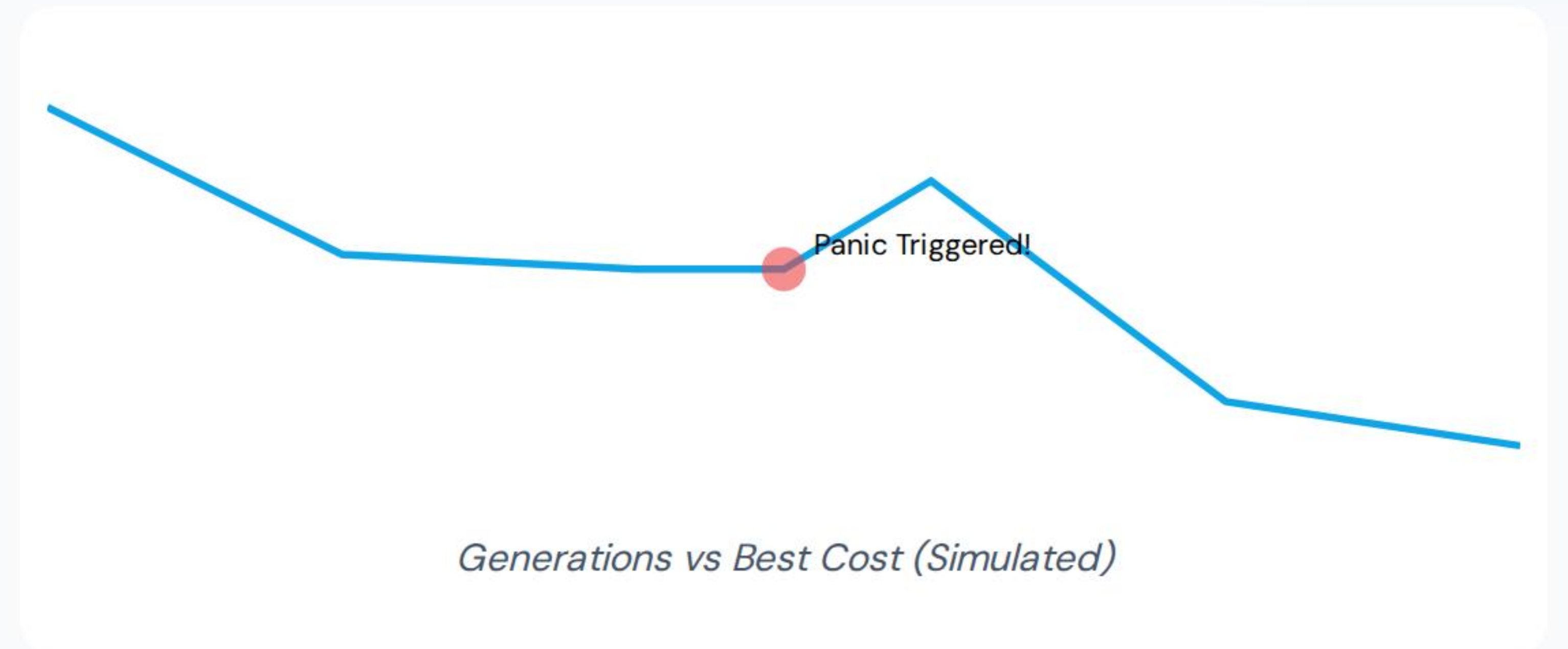
| The "Panic Window"

Escaping Deep Traps

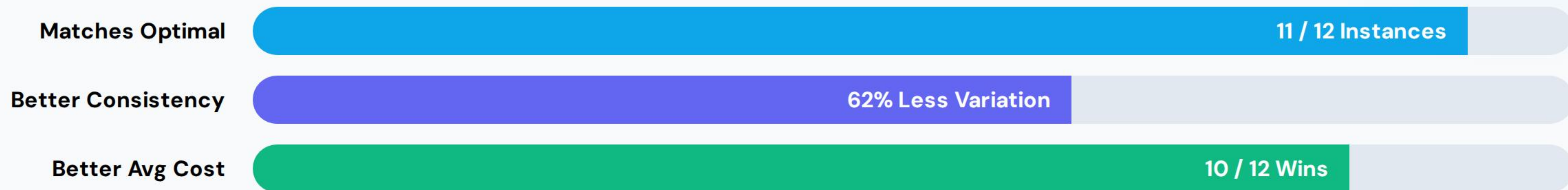
When the algorithm hasn't improved by 0.3% for 300 generations, **Panic** is triggered.

- The best solutions are copied to overwrite the worst ones.
- For 100 generations, we allow "Bad Moves" (blind acceptance).

This "shakes" the system so it can jump out of a trap and find a new, better valley.



Experimental Results



Our Delta RL-MA significantly outperformed the base paper's RL-GA in robustness and reliability across all tested TSPLIB benchmarks.

Thank You!

Questions & Discussion



Implementing Delta RL on Genetic Algorithms for CVRP Optimization

Image Sources



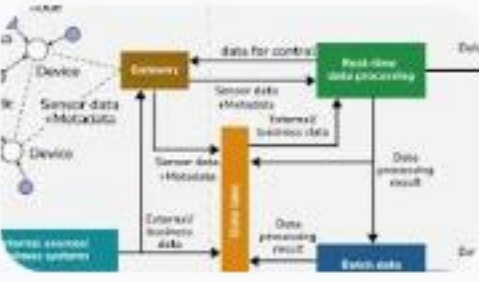
https://www.maersk.com/~media_sc9/maersk/solutions/supply-chain-and-logistics/contract-logistics/warehousing-services/warehousing-services_1232_v3.svg?h=988&w=1232&hash=67D4CC602898FAAC742C3F2F1B49D916

Source: www.maersk.com



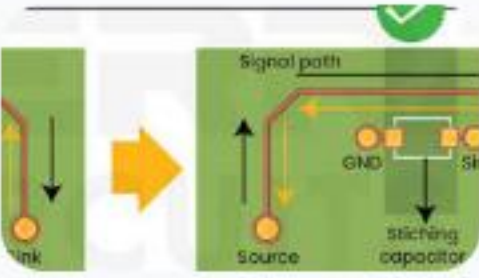
https://miro.medium.com/v2/resize:fit:1200/1*kar7gqRbHR-Q-T8EPCQCVQ.jpeg

Source: medium.com



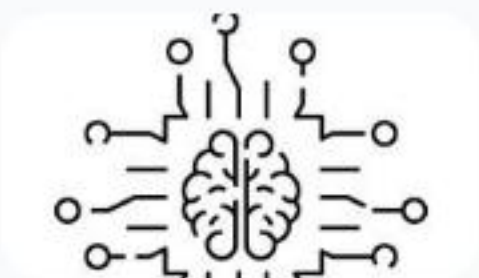
<https://media.geeksforgeeks.org/wp-content/uploads/20250304153519827498/Data-Flow-Diagram.webp>

Source: www.geeksforgeeks.org



<https://www.protoexpress.com/wp-content/uploads/2020/09/routing-over-a-plane-obstruct-high-speed-pcbs.webp>

Source: www.protoexpress.com



https://img.freepik.com/premium-vector/deep-learning-ai-linear-icon-neural-network-with-cogwheels-thin-line-illustration-digital-brain-artificial-intelligence-symbol-vector-isolated-cyber-humanoid-chat_721440-1345.jpg

Source: www.freepik.com



<https://slidemodel.com/wp-content/uploads/4081-01-innovation-project-template-11-870x489.jpg>

Source: slidemodel.com